

BIO DATA



Name : S. Meenakshi Sundaravel
Designation : Assistant Professor (Senior Grade)
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Google Scholar link: <https://scholar.google.com/citations?user=v29NkTwAAAAJ&hl=en&oi=ao>

1. Educational Qualification

Degree	Branch / Specialization	Institute/University	Year of Passing
B.E.	Electrical and Electronics Engineering	Francis Xavier Engineering College, Anna University	2009
M.E	Power Electronics and Drives	Einstein College of Engineering, Anna University	2013
Ph.D	Power Electronics	MEPCO SCHLENK Engineering College, Anna University	Thesis submitted on 29.12.2025

2. Professional/ Industry Experience: (in chronological order)

S.No	Designation	Institution/ Organization	Period	
			From	To
1.	Assistant Professor (Senior Grade)	RAMCO Institute of Technology	01.06.2018	Till date
2.	Assistant Professor	RAMCO Institute of Technology	01.06.2016	31.05.2018
3.	Assistant Professor	St. Mother Theresa Engineering College	01.06.2013	27.05.2016
4.	Lecturer	SCAD College of Engineering and Technology	01.07.2010	01.07.2011

3. Research Interest

Multilevel Inverters

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4 Publications

4.1 JOURNAL PUBLICATIONS

1. Meenakshi Sundaravel, S & Manikandan Bairavan Veerayan 2025, 'Reduced switch single source multilevel inverter topology for renewable photovoltaic system applications', Journal of Power Electronics, vol. 25, no. 3, pp. 440-457, DOI: 10.1007/s43236-024-00910-5, Impact Factor : 1.3. (SCIE)
2. Thangam E, Anandhakumar C, Karthik Prabhu D, Meenakshi Sundaravel S, Vigneshwar A S, "Application Of An Automatic Three-Phase Power Supply Phase Selector In The Medical Ventilator", Journal of Pharmaceutical Negative Results, Vol no. 13, Special Issue 7, 2022, pp. 4818-4823. (Scopus)
3. Mr. S. Meenakshi Sundaravel, Mr. P. Ayyanapparaj, Mr. S. VijayKumar, Mr. S. Yogeshwara, "Solar Panel Cooling System", International Journal of Scientific Research in Science and Technology, Volume 9, Issue 1, ISSN: 2395-6011, 2021, PP. 305-309.

4.2 CONFERENCE PUBLICATIONS

1. S. Meenakshi Sundaravel, R. Chandru, T.R. Alagarsamy Yasvanth, G. Balaraman, "Smart Subsurface Irrigation System Using AI and IoT", International Conference on Exploring AI, IOT, Science and Technology (ICEAIST-24), M.A.M School of Engineering, Trichy, 26.04.2024 to 27.04.2024.
2. S. Meenakshi Sundaravel, V. Sabarisha, R. Alagendra, B. Nikil Dev, "Wireless Underground Sensor Network Path Loss Model for Precision Agriculture", National Conference on Emerging Trends, Science, Engineering and Management, Bharath Niketan Engineering College, Theni, 26.04.2024 to 27.04.2024.
3. Shaji. K. A. Theodore, S. Meenakshi Sundaravel, Kavin K. S., P. T. Subasini, L. Anbarasu, Khaliq Ah, "Performance Analysis of AI Based Interleaved Boost Converter under PV Partial Shaded Conditions", Proceedings of the International Conference on Energy, Materials and Communication Engineering, IEEE, 2023, <https://doi.org/10.1109/ICEMCE57940.2023.10434141>. (Scopus)
4. Meenakshi Sundaravel Sethu Ramachandran, Manikandan Bairavan Veerayan, "Analysis of cascaded H-bridge multilevel inverter using various pulse width modulation strategies", AIP Conference Proceedings, 2023, <https://doi.org/10.1063/5.0162827>. (Scopus)
5. S. Abinayasri, R. Thulasi, S. Meenakshi Sundaravel, "Web Development for Blood Bestow App", International Conference on Recent Innovation in Computer,

Communication and Manufacturing (ICRICCM-2023), AAA college of Engineering and Technology, 12.04.2023.

6. S. Meenakshi Sundaravel, M. Hari Ram, S. Senthilkumar, “Smart Helmet for Bike Riders with Alcohol Detection”, National Conference on Recent Trends in Engineering, Science and management, Kurinji College of Engineering and Technology, 09.06.2022.
7. S. Meenakshi Sundaravel, B.V. Manikandan, “Analysis of Cascaded H-Bridge Multilevel Inverter Using Various Pulse Width Modulation Strategies”, International Conference on Smart Technologies and Applications (ICSTA 2022), Ramco Institute of Technology, 11.03.2022 to 12.03.2022.
8. Mr. S. Meenakshi Sundaravel, Mr. P. Ayyanapparaj, Mr. S. VijayKumar, Mr. S. Yogeshwara, “Solar Panel Cooling System”, International Conference on Advances in Materials, Computing and Communication Technologies ICAMCCT-2021, Annai Vailankanni College of Engineering, AVK Nagar, Kanyakumari, Tamil Nadu, 10th April, 2021.
9. P. Ayyanapparaj, S. Yogeshwara, S. Vijay Kumar, S. Meenakshi Sundaravel, "Smart Automation using Voice Recognition", International Web Conference on Smart Engineering Technologies, Ramco Institute of Technology, 26.06.20 to 27.06.20.
10. M. Gayathiri, M. Kamatchipriya, S. Sujitha, S. Meenakshi Sundaravel, “Solar Panel Cooling System”, National Conference on Computer, Electrical and Electronics Engineering, PSR Rengasamy College of Engineering for Women, 6th March 2020.
11. “PLC based multi-mode cooling of transformer”, Third National Conference on “Intelligence in Electronics and Communication Engineering” NCIECE ’19, RVS College of Engineering and Technology, Coimbatore.
12. “Automatic Voltage Regulation in Power Transmission lines using IoT”, National Conference on “Renewable Energy Systems, Embedded Systems and Robotics” (NCREER - 2018), Kamaraj College of Engineering and Technology, Virudhunagar.
13. “Leakage Detection Monitoring and Control using IOT”, National Conference on Emerging Trends in Electrical, Electronics and Computer Engineering (NCEEECE ’18), JCT College of Engineering and Technology, Coimbatore.
14. “Automatic Detection of Insects in soft drinks using PLC”, National Conference on Innovations in Engineering, Science and Technology – NCIEST’17, Ramco Institute of Technology, Rajapalayam.

4.3 Book/Book chapter publications:

1. A.S. Vigneshwar, S. Meenakshi Sundaravel, A. Arun Kumar, “Challenges of Municipal Solid Waste Management: An Empirical Study”, in the book titled

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“Changing Face of Agriculture, Agri-Business, Agriculture Marketing and Organic Farming – Innovation and Strategies”, Iterative International Publishers, Volume: 01, 2022. ISBN: 978-81-959356-2-8.

5 Funded Projects:

S.No.	Title of the project	Funding Agency	Duration	Amount (Rs.)	Status
1.	Solar based Portable Bipolar Electrocautery Machine	MSME Idea Hackathon 2023	1 Year	Rs. 15 Lakhs	Granted
2.	SUVC – Electric Solar Vehicle Championship ‘20	TRCL	1 Year	Rs. 2 Lakhs	Granted
3.	eBAJA SAEINDIA 2019	Mahindra and SAEINDIA	1 Year	Rs. 1 Lakhs	Granted
4.	E-Bike speed Controllor system	TNSCST	6 Months	Rs. 10,000/-	Applied
5.	Self-Driving Delivery Robot	TNSCST	6 Months	Rs. 10,000/-	Applied
6.	Autonomous Water Quality Monitoring and Oil Containment Robot	eYantra, IIT Bombay	1 Year	Rs. 85,000/-	Applied
7.	Smart Aid (Watch Paired with Glasses)	eYantra, IIT Bombay	1 Year	Rs. 30,000/-	Applied
8.	Rural & Urban Development / Manufacturing / Engineering Technology	Niral Thiruvizha 2.0	6 Months	Rs. 10,000/-	Applied

6 Ph.D. guidance: NIL

Sl.No	Scholar name	University	Topic	Status

7 Short term Courses/Seminars/FDP/Workshop/Conference Attended:

Sl.No	Title of the event	Organized by	Date	
			From	To
1.	FDP on “Green Hydrogen and the Future of Renewable Energy Systems”	ATAL	15.12.2025	20.12.2025
2.	Extramural Lecture Series (ELS) – From Global Engineering Education Initiative to a Local Implementation – Case Turku University of Applied Sciences”	RIT	14.11.2025	-
3.	FDP on “Electric Vehicles: Technology, Power System, Energy Management, Testing and Certification”	ATAL	23.12.2024	28.12.2024

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4.	Innovation Ambassador Training (Reskilling)	MOE's Innovation Cell & AICTE	30.08.2024	30.09.2024
5.	Exploring AI Tools for Effective Academics	RIT	29.07.2024	-
6.	Extramural Lecture Series on "ELS002 - Integrating the CDIO Principles Through Effective Teaching- Learning Activities"	RIT	29.02.2024	-
7.	One day Naan Mudhalvan Niral Thiruvizha – Mentor Training Program	EDII TN, Virudhunagar Collectorate	08.12.2023	-
8.	FDP on Virtual Labs	IITM PALS with NITK Surathkal	15.11.2023	17.11.2023
9.	DST Funded FEDP (Faculty Entrepreneurship Development Programme)	Kalasalingam Academy of Research and Education	14.08.2023	26.08.2023
10.	FDP on "Design of Electric Vehicles using MATLAB Simulink"	RIT	24.07.2023	26.07.2023
11.	"Power System Operational Challenges with Inverter-Based Resources"	IEEE Power Electronics Society	20.06.2023	-
12.	Workshop on "Journal Paper Writing"	RIT	10.02.2023	-
13.	Certified Trainer – O&M Electrical & Instrumentation Technician – Wind Power Plant	NIWE, Vayumitra - NSDC	02.01.2023	13.01.2023
14.	Webinar on "Academic Research and Publishing in High Quality Journals"	Centre for Research, Anna University	20.12.2022	-
15.	Innovation Ambassador Training (Advanced Level)	MOE's Innovation Cell & AICTE	26.09.2022	26.10.2022
16.	Workshop on "Augmented Reality, Virtual Reality & Metaverse Development"	Nan Mudhalvan, Govt. of Tamilnadu	22.09.2022	-
17.	5-day Face-to-Face UHV – II FDP	AICTE	26.07.2022	30.07.2022
18.	Webinar on "How to write an effective seminar/Project proposal for Electrical Engineering"	Kongu Engineering College	26.01.2022	-
19.	Incorporating Universal Human Values in Education	AICTE	20.12.2021	24.12.2021
20.	FDP on Power Electronic Converters and Control for Renewable Power and Micro grid systems	ATAL	14.12.2021	18.12.2021
21.	Innovation Ambassador Training (Foundation Level)	MOE's Innovation Cell & AICTE	30.06.2021	30.07.2021
22.	Webinar on "Implementation of National Educational Policy-2020"	RIT	24.03.2021	-
23.	Workshop on "PLC Programming"	PACR Polytechnic College, DOTE	12.03.2021	21.03.2021

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24.	“C Programming course”	RIT - CSE	06.01.2021	12.02.2021
25.	FDP on “Circuit Design using AutoCAD Electrical”	ICT Academy	25.01.2021	30.01.2021
26.	FDP on “Power Electronics application in Machine Drives and Power System”	ATAL	01.12.2020	05.12.2020
27.	FDP on “Electric Vehicles”	ATAL	2.11.2020	6.11.2020
28.	Course on Online Teaching	IUCEE	July’20	Sep’20
29.	Hands on training on Virtual Lab	IIT PALS with NIT Surathkal	10.08.2020	14.08.2020
30.	FDP on “Role of Faculty in Challenging Situation”	RIT	26.07.20	-
31.	One-week Faculty Development Programme on “Recent Advances in Power Electronics Applications with MATLAB Simulations”	Electronics and ICT Academy, NIT Patna	25.05.2020	30.05.2020
32.	Webinar on “What the best college teachers do”	Ramco Institute of Technology	29.05.2020	-
33.	Webinar on “Matlab Programming and Simulation for Engineering Applications”	Power Innovative Research Consultancy, Chennai	20.05.2020	21.05.2020
34.	Webinar on How to formulate the research problem	Ramco Institute of Technology	13.05.2020	-
35.	Webinar on Effective use of Learning Management System	Ramco Institute of Technology	09.05.2020	-
36.	Faculty Development Training Programme on Technical Advancement in Power System Control, Power Controllers, Drives and E- vehicles	Dr. N.G.P Institute of Technology, Coimbatore. AICTE - IIPC	27.04.2020	01.05.2020
37.	Webinar on Basics of Robotics Simulation	IEEE YESIST12-2020	26.04.2020	27.04.2020
38.	Three day Faculty Development Programme on “Team Building”	ICT Academy, RAMCO Institute of Technology	18.11.2019	20.11.2019
39.	Faculty Development Programme by IIT PALS	Teaching Learning Centre, IIT Madras	12.11.2019	14.11.2019
40.	BAJA SAEINDIA 2019	Mahindra, SAEINDIA, Madhya Pradesh	23.01.2019	27.01.2019
41.	Two day seminar on “Intelligent Electric Power Grid”	RAMCO Institute of Technology	28.12.2018	29.12.2018
42.	TEQIP – III Sponsored Short Term Course on “Recent Trends in Electric Vehicles”	Thiagarajar College of Engineering, Madurai	06.12.2018	08.12.2018
43.	Faculty Development Programme in Entrepreneurship	Ministry of Science and Technology , CED Training Campus, Madurai	12.11.2018	24.11.2018
44.	eyantra Two day workshop on “Introduction to Robotics”	IIT Bombay, CMR Institute of Technology, Bengaluru	05.10.2018	06.10.2018

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45.	Virtual BAJA SAEINDIA 2018	Mahindra, SAEINDIA, Punjab	13.07.2018	14.07.2018
46.	FDTP on “Signals and Systems”	RAMCO Institute of Technology	13.06.2018	18.06.2018
47.	AICTE Sponsored FDP on “Human Values and Professional Ethics”	Ramco Institute of Technology	30.10.2017	12.11.2017
48.	Anna University approved FDTP on “EC6202 - Electronic Devices and Circuits”	KLN College of Engineering and Technology	19.06.2017	25.06.2017
49.	Annual Nodal Centre Conference, Value Virtual Labs, Amrita University	Amrita University	12.05.2017	-
50.	QIP Sponsored one week short Term Course on “Power Electronic Systems and Control”	Pondicherry Engineering College	19.12.2016	23.12.2016
51.	FDP on “Introduction to Research”	Ramco Institute of Technology	18.11.2016	19.11.2016
52.	Anna University approved FDTP on “IC6501 – Control Systems”	Ramco Institute of Technology	04.06.2016	10.06.2016

8. Short term Courses / Seminars /FDP/Workshop/Conference organized:

Sl.No	Title of the event	Organized by	Date	
			From	To
1.	Freshers’ Day 2025	RIT	05.09.2025	-
2.	International Conference on Smart Engineering for Renewable Energy Technologies 2023 (ICSERET - 2023)	RIT	24.03.2023	25.03.2023
3.	Srinivasa Ramanujan Mathematical Competition for School Students – 2022	ISTE	27.01.2023	-
4.	Webinar on “RASPERRY PI”	RIT - ARC	07.10.2022	-
5.	Online Workshop on “Sensors and Displays”	RIT	17.06.2022	18.06.2022
6.	ISTE Student Convention	ISTE	26.03.2022	-
7.	Virtual International Conference on Smart Technologies and Applications (ICSTA 2022)	RAMCO Institute of Technology	11.03.2022	12.03.2022
8.	STTP on “Art of Learning”	RAMCO Institute of Technology	04.12.2021	06.12.2021
9.	Workshop on “TinkerCAD Programming”	RAMCO Institute of Technology	17.09.2021	-
10.	Webinar on “Sensors and Displays”	RAMCO Institute of Technology	26.05.2021	27.05.2021

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11.	Webinar on “Hands On Training: Virtual Laboratory”	RAMCO Institute of Technology	03.11.2020	-
12.	Case Studies In Deep Learning- Hands-On Session Using Google Co-Lab	RAMCO Institute of Technology	16.10.2020	17.10.2020
13.	STTP on “Hands On Training: Virtual Laboratory”	RAMCO Institute of Technology	03.08.2020	08.08.2020
14.	Webinar on “Applied AI and Machine Learning use cases for Enterprises”	RAMCO Institute of Technology	13.06.2020	-
15.	ESVC – Junior Electric Solar Vehicle Championship	ISIE, RIT	02.02.2020	-
16.	PLC Guru in Tech Yuga’19	RAMCO Institute of Technology	31.08.2019	-
17.	Master in You” in Tech Yuga’19	RAMCO Institute of Technology	31.08.2019	-
18.	Science day events by Automation and Robotics club	RAMCO Institute of Technology	28.02.2019	-
19.	Workshop on “Industry V4.0”	RAMCO Institute of Technology	31.08.2018	-
20.	“Master in You” in Tech Yuga’18	RAMCO Institute of Technology	31.08.2018	-
21.	Inauguration of Automation Robotics Club (RIT - ARC)	RAMCO Institute of Technology	28.02.2018	-
22.	“Master in You” in Tech Yuga’17	RAMCO Institute of Technology	12.08.2017	-
23.	“Ardubotics Robotics” - Workshop on Arduino Based Robotics	RAMCO Institute of Technology	28.07.2017	29.07.2017
24.	“Master in You” in Tech Yuga’16	RAMCO Institute of Technology	19.08.2016	-
25.	Training Programme on “NI Certified LabVIEW Associate Developer (CLAD)”	RAMCO Institute of Technology	15.06.2016	21.06.2016

9. Online Courses

Sl.No	Course title	University	Date
1.	Design of Power Electronic Converters	NPTEL	18.05.2025
2.	Data Science Math Skills	Duke University	06.03.2025
3.	eYantra Online Course on Innovation and Entrepreneurship	eYantra MOOC, IIT Bombay	12.11.2024
4.	eLSI: Basics of Embedded Systems and Robotics	eYantra MOOC, IIT Bombay	27.09.2024

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5.	Mathematics for Machine Learning: Multivariate Calculus	Imperial College London	28.06.2024
6.	Fundamentals of Power Electronics	NPTEL	11.05.2024
7.	Battery Comparison, Manufacturing, and Packaging	Arizona State University	19.01.2024
8.	Differential Equations for Engineers	The Hong Kong University of Science and Technology	26.06.2023
9.	Wind Energy	Technical University of Denmark (DTU)	29.05.2023
10.	Mathematics for Machine Learning: Linear Algebra	Imperial College London	30.06.2022
11.	Introduction to Programming with MATLAB	Vanderbilt University	21.12.2021
12.	18.03L: Transfer Functions and the Laplace Transform	MITx	01.10.2020
13.	Introduction to Electronics	Georgia Institute of Technology	08.08.2020
14.	Programming for Everybody (Getting Started with Python)	University of Michigan	11.07.2020
15.	Vector Calculus for Engineers	The Hong Kong University of Technology	22.06.2020
16.	Introduction to Differential Equations	MITx	26.12.2019
17.	Circuits and Electronics 1: Basic circuit analysis	MITx	31.05.2019
18.	Calculus 1C: Coordinate Systems & Infinite Series	MITx	12.05.2019
19.	Differential Equations: Fourier Series and Partial Differential Equations	MITx	21.12.2018
20.	Calculus 1B: Integration	MITx	09.03.2018
21.	Calculus 1A: Differentiation	MITx	10.09.2017
22.	Introduction to Power Electronics: Part 1	University of Colorado Boulder, USA	18.02.2017
23.	ET3034x: Solar Energy	Delft University, USA	30.12.2016

10. Patents filed/awarded:

Sl. No.	Title	Application No.	Filed Date	Status	Inventor(s)

11. Membership in Professional societies

a. ISTE (Life Member)

b. IE(I) (Life Member)

12. Awards, Recognition & Achievements

- Four Anna University rank holders were achieved by the EEE 2020–2024 batch, during my tenure as Class Advisor.
- As a Department Admission Coordinator, achieved 63 (60+3) admissions for the two consecutive years (Batch 2024 – 2028 and Batch 2025 - 2029).
- Delivered an Industrial lecture about “Introduction to Power Electronic Converters and Inverters” for the employees of Rajapalayam Mills Limited on 10.04.2025.
- Recognized as a Certified Trainer for providing training in “O&M Electrical & Instrumentation Technician – Wind Power Plant” to unemployed youths as a part of Vayumitra initiative in collaboration with NSDC.
- Recognized as a Verifier in Yukti – National Innovation Repository (MHRD) and evaluated the idea PoCs, Innovation/Prototypes submitted by our department faculty members and students.
- Won “Best Paper Award” for the paper “Web Development for Blood Bestow App” in the International Conference on Recent Innovation in Computer, Communication and Manufacturing (ICRICCM-2023) conducted by AAA college of Engineering and Technology on 12.04.2023.
- Contributed in department level in achieving 4 star rating for our college in Institution's Innovation Council (IIC) 3.0 rating in 2021.
- Won “Best Paper Award” for the paper entitled “Solar Panel Cooling System” in the International Conference ICAMCCT-2021 held at Annai Vailankanni College of Engineering, Kannyakumari on 10th April, 2021.
- As a faculty coordinator for our college team “Team RITZY”, motivated the students to participate in the main event eBAJA SAEINDIA 2019 organised by Mahindra and SAEINDIA held at Madhyapradesh and got a subsidy of Rs. 1,00,000/- for clearing all the technical rounds.
- Achieved 100% result in the course EE8015 Electric Energy Generation, Utilization and Conservation taught for VII Semester EEE Students in the academic year 2023 – 2024 Odd semester.
- Awarded for producing 100% result in the subject EC8353 Electron Devices and Circuits for the academic year 2019 – 2020 Odd semester.
- Received cash prize and certificate for achieving 100 % result in GE6252 - Basic Electrical and Electronics Engineering for II Semester B.E Mechanical Engineering ‘B’ students in the academic year 2016 – 2017 Even semester.

Any others (Board of studies/Governing Council/Staff Selection/ Reviewer/other duties)

Institute Level:

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- ❖ Innovation Ambassador
- ❖ Automation and Robotics Club (ARC) Coordinator
- ❖ RIT Admission – Documentation Committee Coordinator
- ❖ RIT Admission – School Visit Committee Member
- ❖ Institution Innovation Council (IIC) Coordinator
- ❖ ISO Internal Auditor
- ❖ Vayumitra Training for Unemployed Youths – Trainer and Coordinator
- ❖ Boys Hostel Study Hour Monitoring - Member

Department Level:

- ❖ Domain Coordinator (Power Electronics)
- ❖ Department Admission Coordinator
- ❖ Department Timetable Coordinator
- ❖ NBA Criteria – 6 In-charge
- ❖ NAAC Criteria – 4 In-charge
- ❖ Class Advisor (2020-2024 Batch)
- ❖ Power Electronics Lab In-charge
- ❖ Social Media – LinkedIn Incharge
- ❖ Text book, Note, record notebook Incharge
- ❖ Department Review Meeting Incharge
- ❖ Anna University Affiliation Incharge
- ❖ Virtual Lab Coordinator
- ❖ ICT Academy Coordinator
- ❖ EEE Seminar Hall Incharge